

Fenwick Tree (Binary Indexed Tree)

Step-by-Step Numerical Explanation

1 Given Array

The array is 1-indexed:

$$A = [2, 1, 3, 4, 5, 6, 7, 8]$$

i	1	2	3	4	5	6	7	8
$A[i]$	2	1	3	4	5	6	7	8

2 Fenwick Tree Concept

Each index stores the sum of a specific range:

$$[i - \text{LSB}(i) + 1, i]$$

Where Least Significant Bit (LSB) is defined as:

$$\text{LSB}(i) = i \& (-i)$$

3 LSB Calculation

Index (i)	Binary	LSB(i)
1	0001	1
2	0010	2
3	0011	1
4	0100	4
5	0101	1
6	0110	2
7	0111	1
8	1000	8

4 Building the Fenwick Tree

BIT[1]

$$\text{Range} = [1, 1] \Rightarrow \text{BIT}[1] = 2$$

BIT[2]

$$\text{Range} = [1, 2] \Rightarrow \text{BIT}[2] = 2 + 1 = 3$$

BIT[3]

$$\text{Range} = [3, 3] \Rightarrow \text{BIT}[3] = 3$$

BIT[4]

$$\text{Range} = [1, 4] \Rightarrow \text{BIT}[4] = 2 + 1 + 3 + 4 = 10$$

BIT[5]

$$\text{Range} = [5, 5] \Rightarrow \text{BIT}[5] = 5$$

BIT[6]

$$\text{Range} = [5, 6] \Rightarrow \text{BIT}[6] = 5 + 6 = 11$$

BIT[7]

$$\text{Range} = [7, 7] \Rightarrow \text{BIT}[7] = 7$$

BIT[8]

$$\text{Range} = [1, 8] \Rightarrow \text{BIT}[8] = 2 + 1 + 3 + 4 + 5 + 6 + 7 + 8 = 36$$

5 Final Fenwick Tree Array

$$\text{BIT} = [0, 2, 3, 3, 10, 5, 11, 7, 36]$$

(Index 0 is unused)

6 Prefix Sum Query Example

Find sum of range $[1, 6]$:

$$\text{sum} = \text{BIT}[6] = 11$$

$$i = 6 - \text{LSB}(6) = 6 - 2 = 4$$

$$\text{sum} = 11 + \text{BIT}[4] = 11 + 10 = 21$$

$$i = 4 - 4 = 0 \quad (\text{stop})$$

$\text{Sum}(1, 6) = 21$

7 Point Update Example

Add +3 to index 3:

$$i = 3 \Rightarrow BIT[3] = 3 + 3 = 6$$

$$i = 4 \Rightarrow BIT[4] = 10 + 3 = 13$$

$$i = 8 \Rightarrow BIT[8] = 36 + 3 = 39$$

8 Important Notes

- Supports prefix sum queries efficiently
- Range sum = $\text{prefix}(r) - \text{prefix}(l)$
- Update and query time: $O(\log n)$
- Space complexity: $O(n)$
- Simpler than Segment Tree